

Practice 1

Define a vector with the elements 4,7,3,9,7.

```
v = [4 7 3 9 7]
```

```
v = 1×5  
    4     7     3     9     7
```

Let's append some new elements to v.

```
w = [v 10 11]
```

```
w = 1×7  
    4     7     3     9     7    10    11
```

Address the 2nd element of w.

```
w(2)
```

```
ans =  
    7
```

Address the first 3 elements.

```
w(1:3)
```

```
ans = 1×3  
    4     7     3
```

Address the 2nd and 4th elements.

```
w([2 4])
```

```
ans = 1×2  
    7     9
```

Address the elements with even indices.

```
w(2:2:end)
```

```
ans = 1×3  
    7     9    10
```

Address those elements which are greater than 5.

```
w(w>5)
```

```
ans = 1×5  
    7     9     7    10    11
```

Define a matrix.

```
r1 = [1 3 5];  
r2 = [4 6 7];  
M = [r1 ; r2]
```

```
M = 2x3
    1    3    5
    4    6    7
```

Address the 1st row.

```
M(1, :)
```

```
ans = 1x3
    1    3    5
```

Create a cell object (list).

```
c = {'a',4,[2 4]}
```

```
c = 1x3 cell
```

	1	2	3
1	'a'	4	[2,4]

Address the 3rd object.

```
c{3}
```

```
ans = 1x2
    2    4
```

Structure: labeled list

```
s = struct('X',[1 2],'Y',4)
```

```
s = struct with fields:
    X: [1 2]
    Y: 4
```

Address the X variable.

```
s.X
```

```
ans = 1x2
    1    2
```

Table is the dataframe in Matlab.

```
x = [1 4 5]'; % ' is transpose
y = [5 6 9]';
t = table(x,y)
```

```
t = 3x2 table
```

	x	y
1	1	5
2	4	6
3	5	9

```
t.x
```

```
ans = 3×1
      1
      4
      5
```

```
t(:,1) % same as the previous
```

```
ans = 3×1 table
```

	x
1	1
2	4
3	5

Data import.

```
% Import data from text file
pwd
```

```
ans =
'/home/nosy/Asztal/git/edu-related/_app_stat_lab/lab-7'
```

```
ustemp = readtable("data/ustemp.txt");
```

```
% Display results
ustemp
```

```
ustemp = 56×4 table
```

	City	JanTemp	Lat	Long
1	'Mobile, AL'	44	31.2000	88.5000
2	'Montgomery, AL'	38	32.9000	86.8000
3	'Phoenix, AZ'	35	33.6000	112.5000
4	'Little Rock, AR'	31	35.4000	92.8000
5	'Los Angeles, CA'	47	34.3000	118.7000
6	'San Francisco, CA'	42	38.4000	123
7	'Denver, CO'	15	40.7000	105.3000
8	'New Haven, CT'	22	41.7000	73.4000
9	'Wilmington, DE'	26	40.5000	76.3000
10	'Washington, DC'	30	39.7000	77.5000
11	'Jacksonville, FL'	45	31	82.3000
12	'Key West, FL'	65	25	82
13	'Miami, FL'	58	26.3000	80.7000
14	'Atlanta, GA'	37	33.9000	85
15	'Boise, ID'	22	43.7000	117.1000

	City	JanTemp	Lat	Long
16	'Chicago, IL'	19	42.3000	88
17	'Indianapolis, IN'	21	39.8000	86.9000
18	'Des Moines, IA'	11	41.8000	93.6000
19	'Wichita, KS'	22	38.1000	97.6000
20	'Louisville, KY'	27	39	86.5000
21	'New Orleans, LA'	45	30.8000	90.2000
22	'Portland, ME'	12	44.2000	70.5000
23	'Baltimore, MD'	25	39.7000	77.3000
24	'Boston, MA'	23	42.7000	71.4000
25	'Detroit, MI'	21	43.1000	83.9000
26	'Minneapolis, MN'	2	45.9000	93.9000
27	'St. Louis, MO'	24	39.3000	90.5000
28	'Helena, MT'	8	47.1000	112.4000
29	'Omaha, NE'	13	41.9000	96.1000
30	'Concord, NH'	11	43.5000	71.9000
31	'Atlantic City, NJ'	27	39.8000	75.3000
32	'Albuquerque, NM'	24	35.1000	106.7000
33	'Albany, NY'	14	42.6000	73.7000
34	'New York, NY'	27	40.8000	74.6000
35	'Charlotte, NC'	34	35.9000	81.5000
36	'Raleigh, NC'	31	36.4000	78.9000
37	'Bismarck, ND'	0	47.1000	101
38	'Cincinnati, OH'	26	39.2000	85
39	'Cleveland, OH'	21	42.3000	82.5000
40	'Oklahoma City, OK'	28	35.9000	97.5000
41	'Portland, OR'	33	45.6000	123.2000
42	'Harrisburg, PA'	24	40.9000	77.8000
43	'Philadelphia, PA'	24	40.9000	75.5000
44	'Charleston, SC'	38	33.3000	80.8000
45	'Nashville, TN'	31	36.7000	87.6000
46	'Amarillo, TX'	24	35.6000	101.9000
47	'Galveston, TX'	49	29.4000	95.5000
48	'Houston, TX'	44	30.1000	95.9000

	City	JanTemp	Lat	Long
49	'Salt Lake City, UT'	18	41.1000	112.3000
50	'Burlington, VT'	7	45	73.9000
51	'Norfolk, VA'	32	37	76.6000
52	'Seattle, WA'	33	48.1000	122.5000
53	'Spokane, WA'	19	48.1000	117.9000
54	'Madison, WI'	9	43.4000	90.2000
55	'Milwaukee, WI'	13	43.3000	88.1000
56	'Cheyenne, WY'	14	41.2000	104.9000

```
% January temperatures in Fahrenheit
ustemp.JanTemp
```

```
ans = 56×1
    44
    38
    35
    31
    47
    42
    15
    22
    26
    30
     ⋮
```

Convert the JanTemp from Fahrenheit if

- $32\text{ }^{\circ}\text{F} = 0\text{ }^{\circ}\text{C}$
- $212\text{ }^{\circ}\text{F} = 100\text{ }^{\circ}\text{C}$

```
tempc = (ustemp.JanTemp - 32) / (212-32) * 100
```

```
tempc = 56×1
    6.6667
    3.3333
    1.6667
   -0.5556
    8.3333
    5.5556
   -9.4444
   -5.5556
   -3.3333
   -1.1111
     ⋮
```

Mean temperature, standard deviation, median, quartiles, range.

```
mean(tempc)
```

```
ans =  
-3.0456
```

```
std(tempc)
```

```
ans =  
7.4332
```

```
median(tempc)
```

```
ans =  
-4.1667
```

```
quantile(tempc, [.25 .75])
```

```
ans = 1×2  
-7.5000    0.8333
```

```
range(tempc)
```

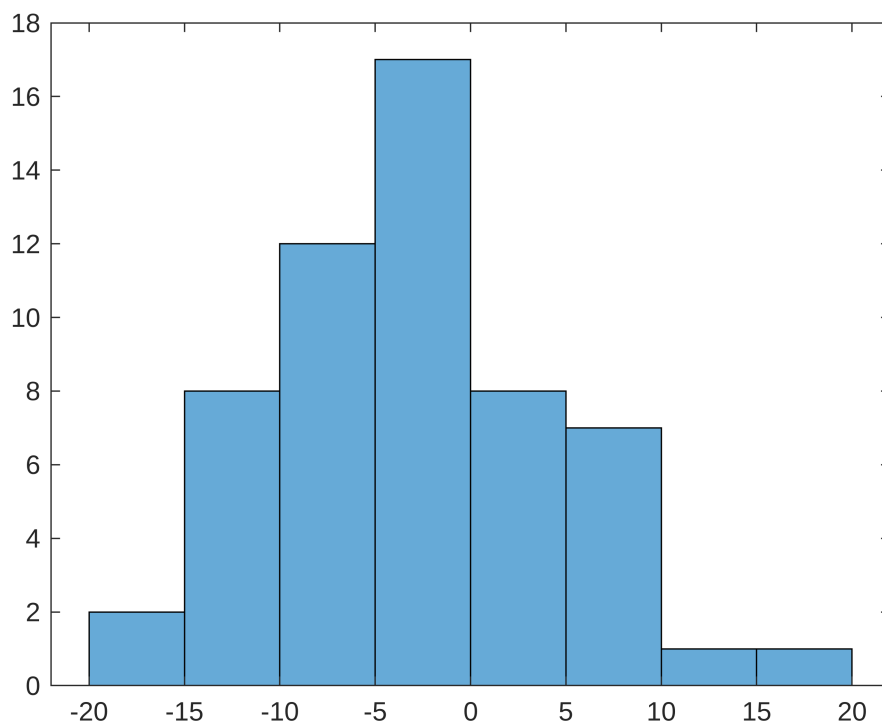
```
ans =  
36.1111
```

Distribution.

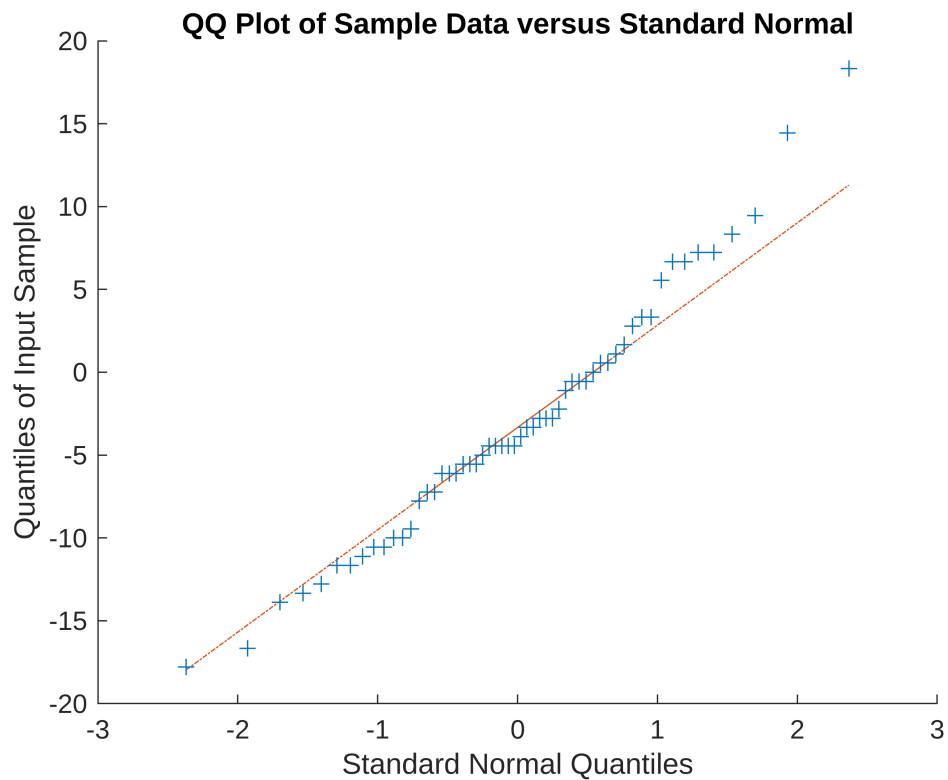
```
skewness(tempc)
```

```
ans =  
0.5021
```

```
histogram(tempc)
```



```
qqplot(tempc)
```



Exercise: A cyclist in a race had the following time results in 4 stages.

1. stage: 120km, 245 min
2. stage: 130km, 258 min
3. stage: 110km, 239 min
4. stage: 120km, 249 min

What was his average speed (km/h) in each stages?

```
l = [120 130 110 120];  
t = [245 258 239 249]/60;  
speed = l./t;  
disp(speed)
```

```
29.3878    30.2326    27.6151    28.9157
```

Exercise: The daily returns of Bitcoin were +1%,-5%,+2%,-3%.

Calculate the average return.

$$\text{geomean}(x) = \sqrt[n]{\prod_{i=1}^n x_i}$$

```
geomean([1.01 0.95 1.02 0.97])
```

```
ans =  
0.9871
```

Exercise: A swimmer swims one lap of freestyle at 3km/h, then one lap of breaststroke at 2km/h. What is the average speed?

$$\text{harmmean}(x) = \left(\frac{1}{n} \sum_{i=1}^n \frac{1}{x_i} \right)^{-1}$$

```
harmmean([2 3])
```

```
ans =  
2.4000
```

Example, random number generator.

Function: $D(x) = 4x(1 - x)$.

Iteration yields a sequence which has the same behaviour as a random sequence from arcsine distribution with

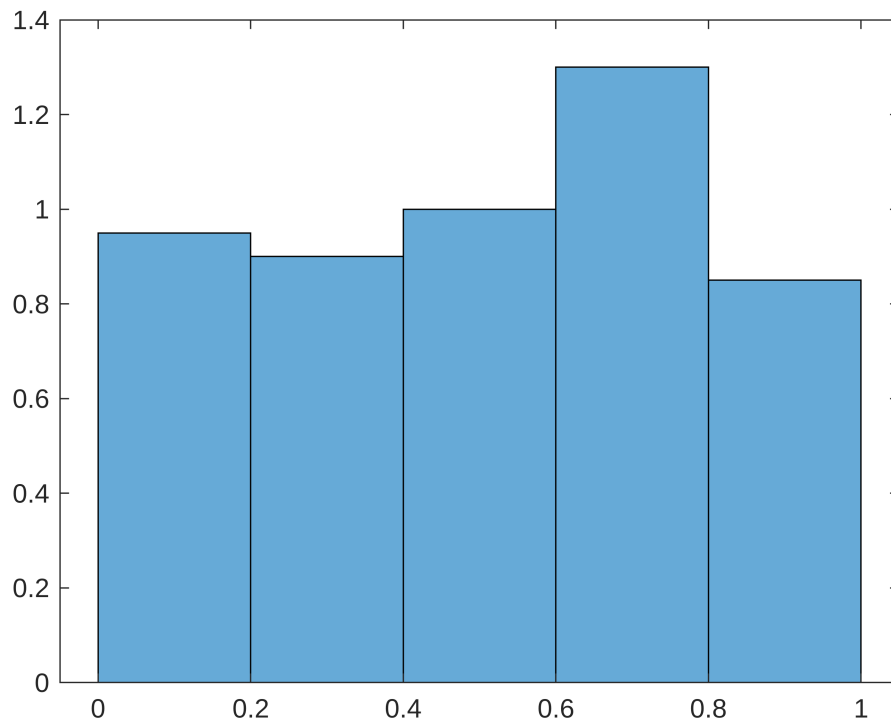
$$F(x) = \frac{\arcsin(2x - 1)}{\pi} + \frac{1}{2}$$

Iterations:

$$x_0 = 1/3, x_{n+1} = D(x_n), y_n = F(x_n).$$

Check the histogram every hundredth element of y .

```
x = zeros(1,10000);  
x(1) = 1/3;  
  
for i = 2:10000  
    x(i) = 4*x(i-1)*(1-x(i-1));  
end  
y = asin(2*x-1)/pi+.5;  
histogram(y(1:100:end), 'Normalization', 'pdf')
```

Distributions in matlab.

```
normcdf(-50,20,50) % ~N(20,50^2)
```

```
ans =  
0.0808
```

```
normpdf(-50,20,50)
```

```
ans =  
0.0030
```

```
norminv(0.5,0,1) % 0.5-quantile of N(0,1)
```

```
ans =  
0
```

```
normrnd(0,1,[1 10]) % [1 10] sample from N(0,1)
```

```
ans = 1×10  
1.0933    1.1093   -0.8637    0.0774   -1.2141   -1.1135   -0.0068    1.5326 ...
```

```
[m sd]=normfit([0.2 -0.3 1.1]) % estimate from the sample
```

```
m =  
0.3333  
sd =  
0.7095
```

```
[m v] = unifstat(0,1) % theoretical mean and variance
```

```
m =  
0.5000  
v =  
0.0833
```

Plot the pdf of $N(10, 20^2)$.

```
x = linspace(-50, 70); % domain  
y = normpdf(x, 10, 20);  
plot(x, y)
```

Plot the pdf of $N(10, 30^2)$ on the previous figure.

```
hold on  
z = normpdf(x, 10, 30);  
plot(x, z)  
hold off  
legend('N(10, 202)', 'N(10, 302)')
```

